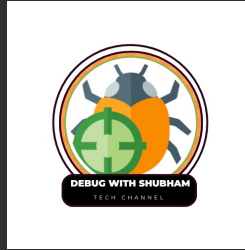


Debug With Shubham Notes



<https://www.youtube.com/@DebugWithShubham>

[linkedin.com/in/debugwithshubham](https://www.linkedin.com/in/debugwithshubham)

[instagram.com/debugwithshubham](https://www.instagram.com/debugwithshubham)

t.me/debugwithshubham

topmate.io/debugwithshubham



NQT 2025



2-3 Days Interview Plan & Que & Ans

Ninja Role 2025

DAY 1: Core Technical (Coding)



Time Needed: 1 0 hours



Focus: Logic building, basic DSA



Patterns (star, number, pyramid)



Array (frequency, duplicates, reverse)



String (palindrome, anagram, character count)

Practice Sites:

[LeetCode](#)

[GeeksforGeeks](#)



Target: Solve 1 0 - 1 5 questions that match past TCS patterns.

Day 2- SQL + DMBS + Core Sub

✓ SELECT, WHERE, GROUP BY, HAVING

✓ JOINS – especially INNER JOIN, LEFT JOIN

◆ CS Basics:

✓ OOP Concepts (Encapsulation, Inheritance, Polymorphism, abstraction)

✓ DBMS (Normalization, Keys)

✓ OS basics (Deadlock, Paging)

✓ Networking (TCP/IP, HTTP vs HTTPS)

Day 3- HR & Managerial Questions + Project

Tell me about yourself

Why TCS / Ninja role?

Strengths and Weaknesses

Describe a project (pick 1 from your resume)

What if you're rejected?

➡ Tip: Practice in front of a mirror or record yourself

Project:

Goal/Purpose

Technologies Used

Your Role

Challenges & Solutions

Output / Result

Coding Question

1-Reverse a String

Input: "hello"

Output: "olleh"

2-Check if a number is Prime

Input: 17

Output: Prime

3-Find the Factorial of a Number

Input: 5

Output: 120

4-Fibonacci Series up to N terms

Input: 5

Output: 0 1 1 2 3

5- Palindrome String Check

Input: "madam"

Output: Palindrome

Find the Largest Element in an Array

Input: [3, 7, 2, 5]

Output: 7

Sum of Digits of a Number

Input: 123

Output: 6

Print Star Pattern (Right Triangle)

Input: 4

Output:

```
*  
* *  
* * *  
* * * *
```

Check Armstrong Number

Input: 153

Output: Armstrong

$(1^3 + 5^3 + 3^3 = 153)$

Count Frequency of Each Character

Input: "aabbcc"

Output: a:2, b:2, c:1

BONUS (Asked in TCS NQT Interview Frequently)

- ◆ Find the Second Largest Number in an Array
- ◆ Remove Duplicates from an Array
- ◆ Find GCD of Two Numbers
- ◆ Find Missing Number in 1-N Sequence
- ◆ Swap Two Numbers Without Temp Variable

SQL

1-Select All Columns from a Table

```
SELECT * FROM Employees;
```

2-Find the Second Highest Salary

```
SELECT MAX(Salary)
FROM Employees
WHERE Salary < (SELECT MAX(Salary) FROM Employees);
```

3-Find Employees from a Specific Department

```
SELECT *
FROM Employees
WHERE Department = 'IT';
```

4-Count the Number of Employees in Each Department

```
SELECT Department, COUNT(*)
FROM Employees
GROUP BY Department;
```

5-Display Total Salary by Department

```
SELECT Department, SUM(Salary)
FROM Employees
GROUP BY Department;
```


6-JOIN Two Tables (INNER JOIN)

```
SELECT e.Name, d.DeptName  
FROM Employees e  
INNER JOIN Departments d ON e.DeptID = d.DeptID;
```

7-Find Employees Who Don't Have a Manager

```
SELECT Name  
FROM Employees  
WHERE ManagerID IS NULL;
```

8-Find Duplicate Employee Names

```
SELECT Name, COUNT(*)  
FROM Employees  
GROUP BY Name  
HAVING COUNT(*) > 1;
```

9-Find Employees Hired in the Last 6 Months

```
SELECT *  
FROM Employees  
WHERE HireDate >= DATEADD(MONTH, -6, GETDATE());
```

10-Subquery to Get Employee with Max Salary

```
SELECT *  
FROM Employees  
WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

BONUS Quick Topics:

DISTINCT, ORDER BY, LIMIT

IN, BETWEEN, LIKE, IS NULL

Difference between WHERE and HAVING

Types of joins (LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN)

DBMS

1-What is DBMS?

A Database Management System is software that manages databases and allows CRUD operations (Create, Read, Update, Delete).

2-What is the difference between DBMS and RDBMS?

DBMS	RDBMS
-----	-----
No relationships	Maintains relations (tables)
No keys required	Uses Primary and Foreign Keys
Less secure	More secure

3-What are Primary and Foreign Keys?

Primary Key uniquely identifies each record (e.g., EmpID)

Foreign Key links one table to another (e.g., DeptID in Employees referencing Departments)

4-What is Normalization?

Normalization is the process of organizing data to reduce redundancy and improve integrity.

1NF: No repeating groups

2NF: No partial dependency

3NF: No transitive dependency

5-What are the types of SQL commands?

DDL: CREATE, ALTER, DROP

DML: INSERT, UPDATE, DELETE

DCL: GRANT, REVOKE

TCL: COMMIT, ROLLBACK

DQL: SELECT

6-What is a Transaction?

A transaction is a single unit of work with ACID properties:

Atomicity

Consistency

Isolation

Durability

Example: Transferring money between bank accounts.

7-What is a View?

A view is a virtual table based on the result of a SQL query.

Difference Between WHERE and HAVING

WHERE	HAVING
-----	-----
Filters rows	Filters groups
Used before GROUP BY	Used after GROUP BY

8-What is Indexing?

Indexing improves search speed on large tables.

Like the index of a book — makes finding faster.

9- Difference: DELETE vs TRUNCATE vs DROP

DELETE: removes rows (can rollback)

TRUNCATE: removes all rows (no rollback)

DROP: deletes the entire table

Bonus Topics:

Candidate Key vs Super Key

E-R Model (Entity, Attributes, Relationships)

Types of Relationships (1:1, 1:N, M:N)

JOIN vs Subquery

Denormalization